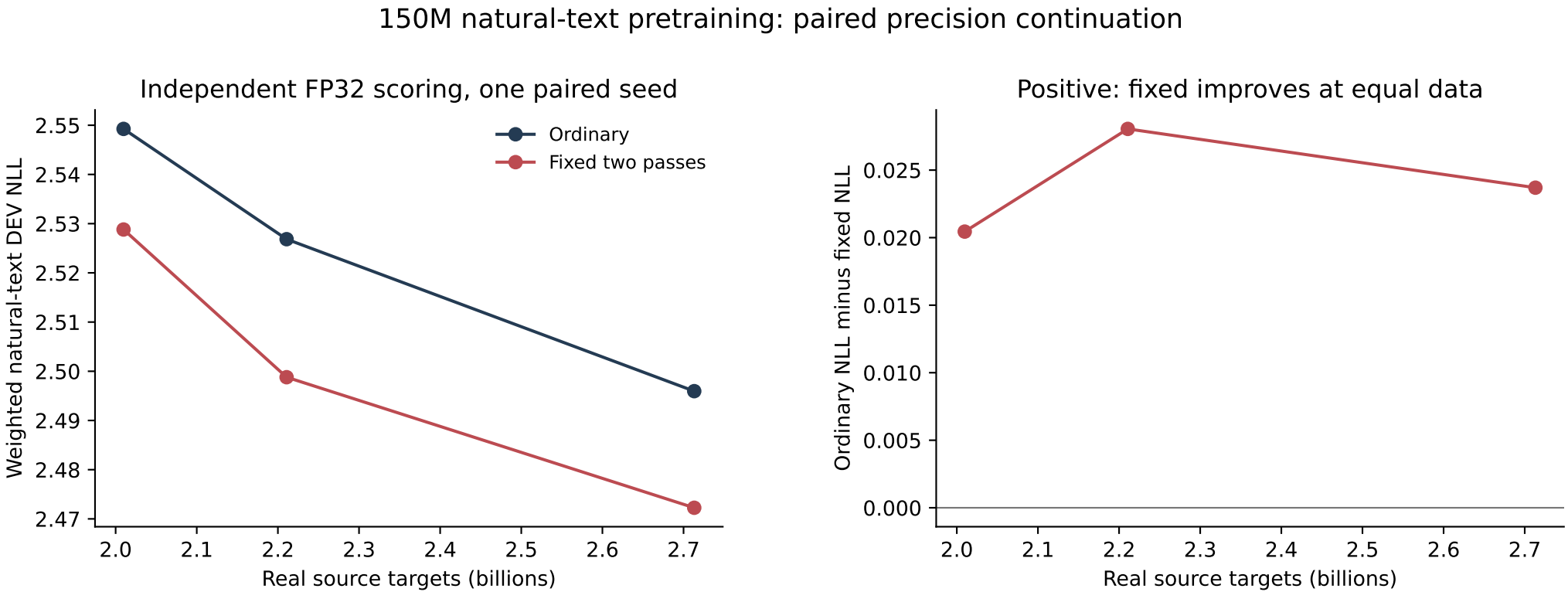
150M natural-text pretraining: paired precision continuation
Independent FP32 scoring, one paired seed
Ordinary
Fixed two passes
Weighted natural-text DEV NLL
2.55
2.54
2.53
2.52
2.51
2.50
2.49
2.48
2.47
2.0
2.1
2.2
2.3
2.4
2.5
2.6
2.7
Real source targets (billions)
Positive: fixed improves at equal data
Ordinary NLL minus fixed NLL
0.025
0.020
0.015
0.010
0.005
0.000
2.0
2.1
2.2
2.3
2.4
2.5
2.6
2.7
Real source targets (billions)
FP32 training starts at2.010B after BF16 history. Circles: stable LR; diamonds: completed decay.
One reused seed, consumed DEV; no seed intervals. Fixed uses1.692× estimated matmuls. No efficiency claim.